OTT Content Viewership Analysis - ShowTime

A Data-Driven Approach to Optimizing First-Day Content Performance

Prepared for: *ShowTime Management Team*
Prepared by: *Twinkle Verma*
Date: *5th April,2026*

1. Executive Summary

This report analyzes the key factors influencing first-day content viewership on the ShowTime OTT platform using a data-driven approach.

A multiple linear regression model was developed to evaluate the impact of platform traffic, audience engagement, marketing efforts, release timing, and external competition on content performance.

The analysis identifies **pre-release audience engagement (trailer views) and platform traffic (visitors) as the most significant drivers of content success**, as both demonstrate a strong positive relationship with first-day viewership. In contrast, the presence of **major sports events negatively impacts performance**, indicating that competing events divert audience attention.

Additionally, **release timing, including day of the week and seasonal trends, has a moderate influence on viewership**, while **ad impressions and genre show limited impact**, suggesting that marketing effectiveness depends more on engagement quality than the volume of promotional efforts.

Overall, the findings emphasize that **engagement-driven strategies and optimized release planning are critical to maximizing first-day content performance**. These insights enable ShowTime to make informed, data-driven decisions to enhance audience engagement and achieve more consistent content success.

2. Business Problem

ShowTime aims to maximize the success of its content by improving first-day viewership, which serves as a key performance indicator for content launches.

However, content performance is influenced by multiple factors, including audience engagement, platform traffic, marketing activities, release timing, and external competition.

The objective of this analysis is to identify the key drivers of first-day content viewership and provide actionable insights to support data-driven decision-making.

3. Data Understanding

The dataset consists of 1000 observations and 8 variables, including both numerical and categorical features.

The target variable is:

- **views_content** (first-day content views in millions)

Key predictor variables include:

- **Visitors** – platform traffic
- **Ad Impressions** – marketing exposure
- **Trailer Views** – audience engagement
- **Genre** – content type
- **Major Sports Event** – external competition
- **Day of Week** – release timing
- **Season** – seasonal trends

The dataset was checked for missing values and duplicates, and no issues were found, indicating good data quality for analysis.

4. Methodology

A structured analytical approach was followed:

- Data validation and cleaning
- Exploratory Data Analysis (EDA) to understand patterns and distributions
- Feature-level analysis to identify relationships
- Model building using **Multiple Linear Regression**
- Interpretation of model results to derive business insights

This approach enables both quantitative analysis and meaningful business interpretation of the results.

5. Exploratory Data Analysis

5.1 Distribution of Content Views

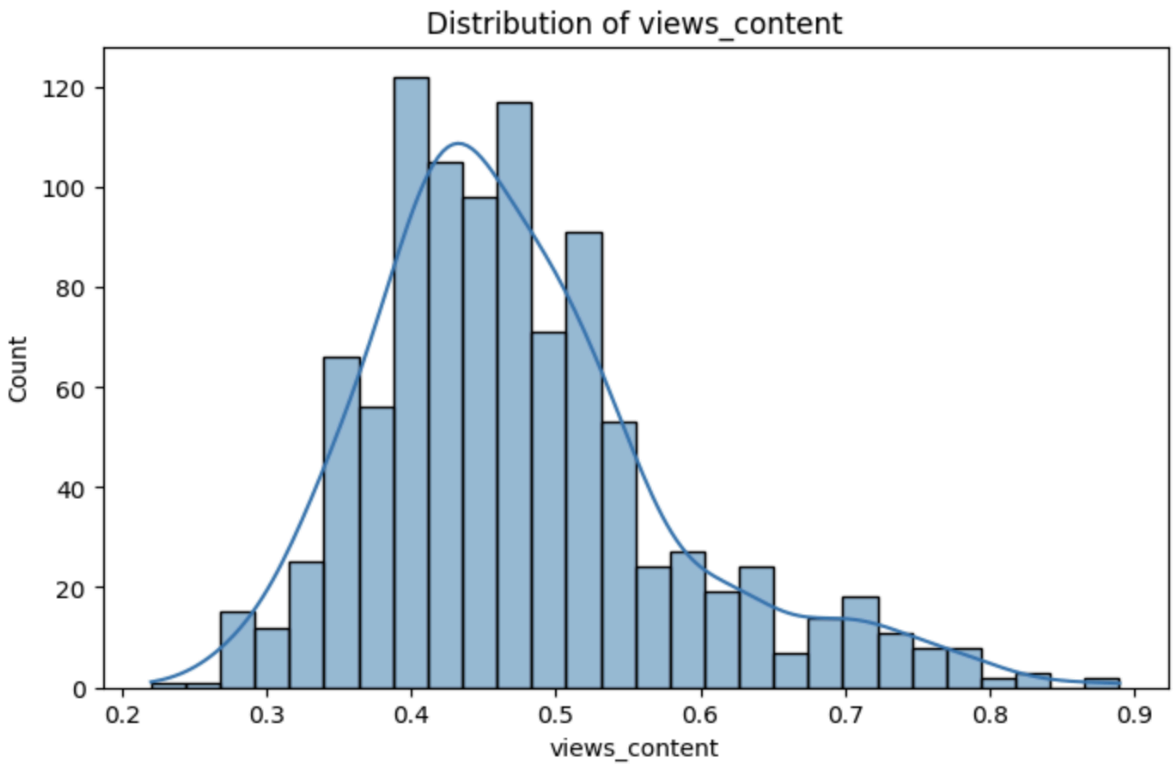

Figure 1: Distribution of First-Day Content Views

The distribution of first-day content views is moderately right-skewed, indicating that while most content performs within a mid-range, a smaller number of titles achieve significantly higher viewership. This suggests the presence of high-performing outliers, highlighting the importance of identifying factors that drive exceptional content success.

5.2 Genre-wise Distribution

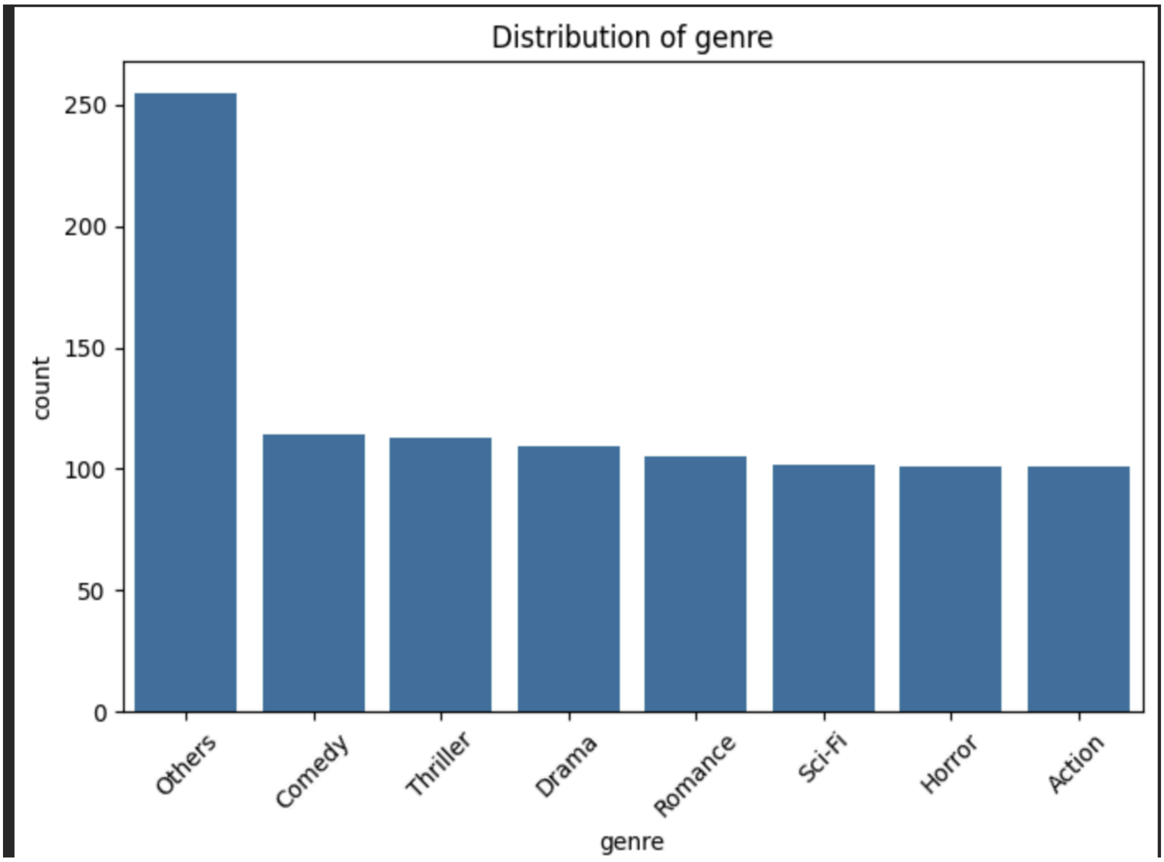

Figure 2: Genre Distribution

The distribution of content across genres appears relatively balanced, with no single genre overwhelmingly dominating the dataset. This indicates that content production is diversified across genres, and performance is unlikely to be driven solely by content type, reinforcing the need to focus on other factors such as engagement and timing.

5.3 Impact of Day of Week

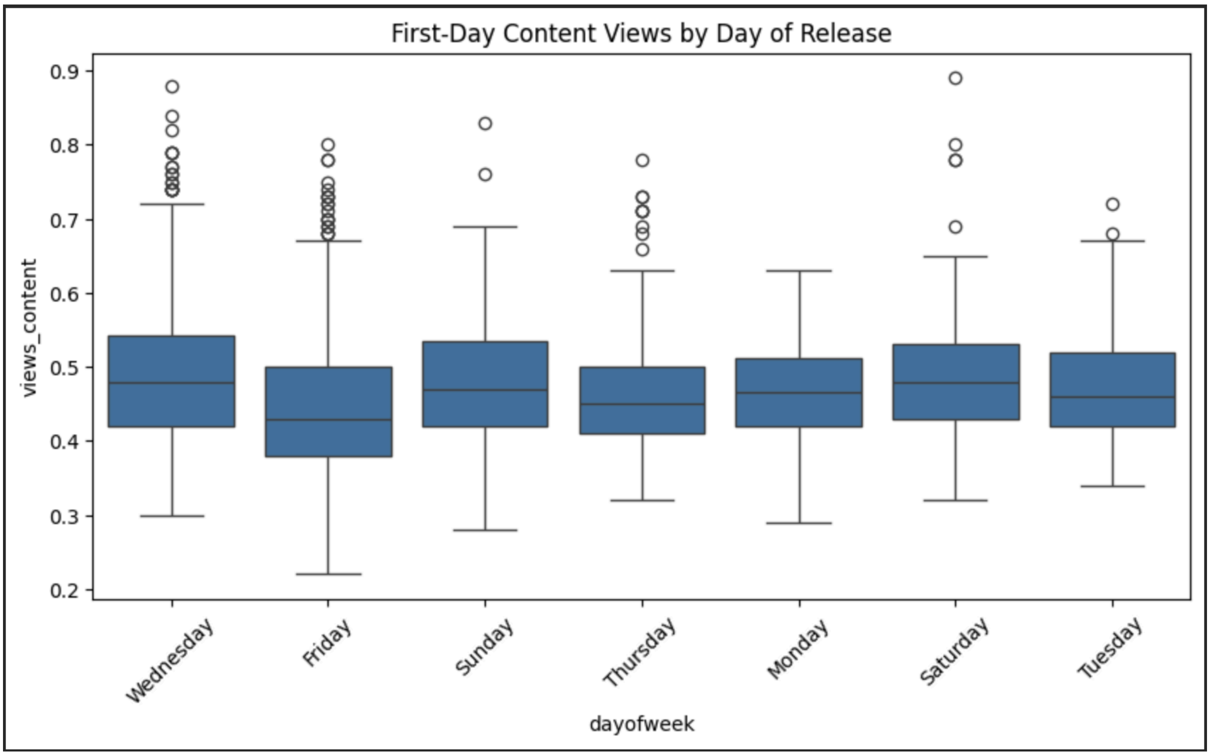

Figure 3: Content Views by Day of Week

Content released during weekends, particularly Saturday and Sunday, shows relatively higher median viewership compared to weekdays. This suggests that audience availability and engagement are higher during weekends, making them more favorable for content releases.

5.4 Seasonal Trends

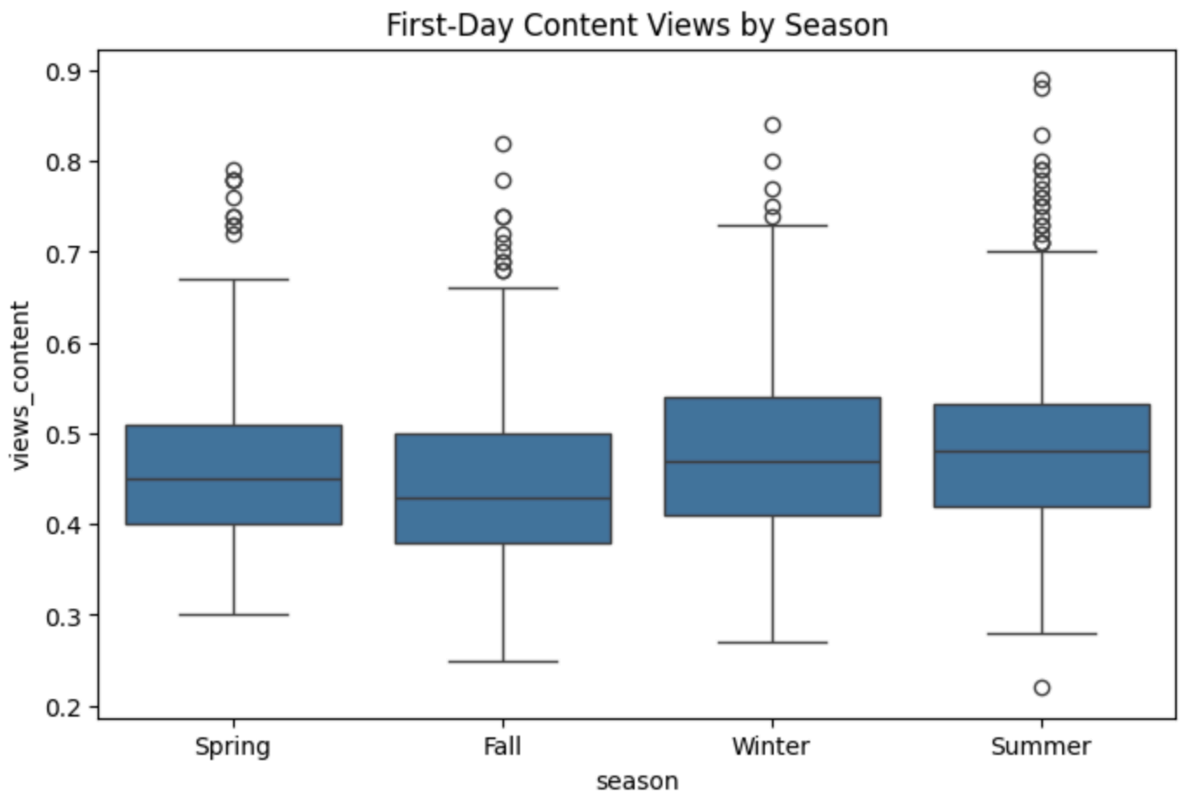

Figure 4: Content Views by Season

Content released during Summer and Winter seasons demonstrates relatively higher median performance compared to other seasons. This indicates that audience activity may be higher during these periods, making them strategically important windows for major content launches.

5.5 Trailer Views vs Content Views

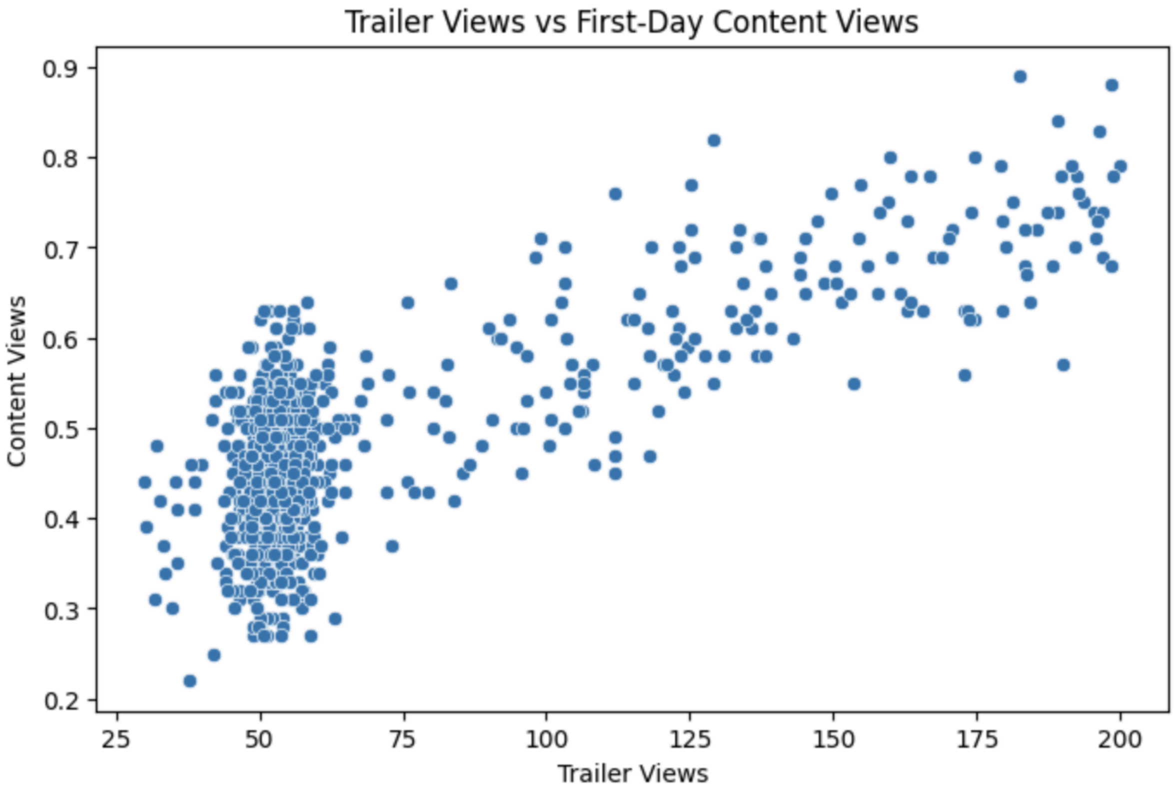

Figure 5: Relationship between Trailer Views and Content Views

A strong positive relationship is observed between trailer views and first-day content views, indicating that higher pre-release audience engagement consistently leads to improved content performance. This highlights the critical role of trailer engagement as a leading indicator of success.

This pattern suggests that while average performance remains stable, a small proportion of content drives disproportionately high viewership, indicating opportunities to replicate high-performing content strategies.

6. Model Building and Results

A multiple linear regression model was developed to assess the impact of key variables on first-day content viewership.

The model provides a structured and quantitative understanding of how platform activity, audience engagement, marketing efforts, and external factors influence content performance.

Key Findings

- **Trailer Views have a strong positive impact**
  Trailer views emerge as the most significant predictor of content performance, indicating that higher pre-release audience engagement directly translates into increased first-day viewership.
- **Visitors have a strong positive impact**
  Platform traffic significantly contributes to content visibility, with higher visitor volumes leading to improved content reach and performance.
- **Major Sports Events have a negative impact**
  The occurrence of major sports events reduces viewership, suggesting that competing events divert audience attention away from newly released content.
- **Day of Week and Season have a moderate impact**
  Temporal factors, including release timing and seasonal trends, show a noticeable influence on performance, highlighting the importance of strategic scheduling.
- **Ad Impressions have limited impact**
  Advertising exposure alone does not significantly drive viewership, indicating that marketing effectiveness depends more on engagement quality than volume.
- **Genre has minimal impact**
  Content type does not significantly influence first-day performance, suggesting that engagement and platform dynamics play a more critical role

Model Interpretation

Overall, the analysis indicates that **audience engagement and platform reach are the primary drivers of content success**, while external competition and timing factors play a supporting role.

These findings suggest that optimizing pre-release engagement strategies and carefully planning release timing can significantly enhance first-day content performance.

The model demonstrates a strong explanatory capability, with key variables showing meaningful influence on content performance. The results are directionally consistent and provide reliable insights into the factors driving first-day viewership.

The model explains a substantial portion of variance in content viewership, indicating a strong overall fit.

Model Assumptions Validation

The assumptions of linear regression were evaluated to ensure the reliability of the model.

- Linearity: The relationship between predictor variables and content viewership is approximately linear, as observed in scatter plots.
- Multicollinearity: Variance Inflation Factor (VIF) values are within acceptable limits, indicating no significant multicollinearity among predictors.
- Homoscedasticity: Residual plots show constant variance, suggesting that the error terms are evenly distributed.
- Normality of Residuals: The residual distribution appears approximately normal.

These results confirm that the model satisfies key statistical assumptions and is suitable for interpretation and decision-making.

Model Performance Metrics

The regression model demonstrates strong predictive performance:

- **R-squared: 0.77**
- **Mean Absolute Error (MAE): 0.04**
- **Root Mean Squared Error (RMSE): 0.05**

The R-squared value indicates that approximately 77% of the variation in first-day content viewership is explained by the model.

The low MAE and RMSE values suggest that prediction errors are minimal, indicating reliable model performance.

7. Key Business Insights

- Audience engagement (trailer views) is the **most critical factor** influencing content success.
- Platform traffic plays a significant role in determining visibility and reach.
- External competition, such as major sports events, negatively impacts viewership.
- Release timing (day and season) has a noticeable but moderate effect.
- High marketing spend does not guarantee success without effective audience engagement.
- Pre-release audience engagement demonstrates a stronger influence on content performance compared to traditional marketing exposure, indicating that engagement quality outweighs promotional volume.

8. Recommendations

- Increase focus on **pre-release trailer engagement strategies**
- Schedule releases during **low-competition periods**
- Optimize release timing based on **high-performing days and seasons**
- Improve **content discoverability on the platform**
- Shift focus from volume-based advertising to **engagement-driven marketing**

9. Limitations

- The model assumes linear relationships between variables
- Qualitative factors such as content quality are not included
- Results are based on historical data and may not fully predict future trends

10. Conclusion

This analysis identifies the key drivers influencing first-day content viewership on the ShowTime platform through a data-driven approach.

The findings highlight that **audience engagement and platform traffic are the most significant contributors to content success**, while external competition, particularly major sports events, negatively impacts performance. Additionally, **release timing plays a meaningful role**, whereas ad impressions and genre have limited influence on viewership outcomes.Overall, the results emphasize that **success is driven more by effective audience engagement and strategic release planning than by the volume of marketing efforts or content type**.

By leveraging these insights, ShowTime can adopt a more targeted and data-driven approach to content releases, enabling improved audience engagement and more consistent content performance.